

DEPARTMENT OF CSE - DS III YEAR I TERM

PROJECT PRESENTATION

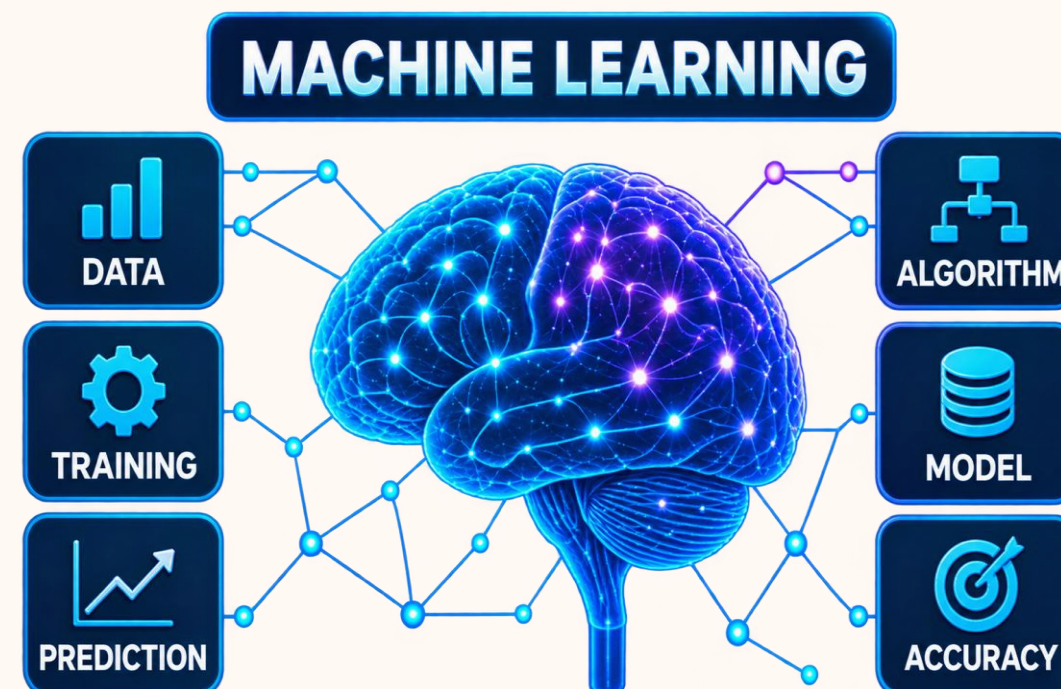
MACHINE LEARNING

Loan Approval Prediction using Logistic Regression and Decision Tree

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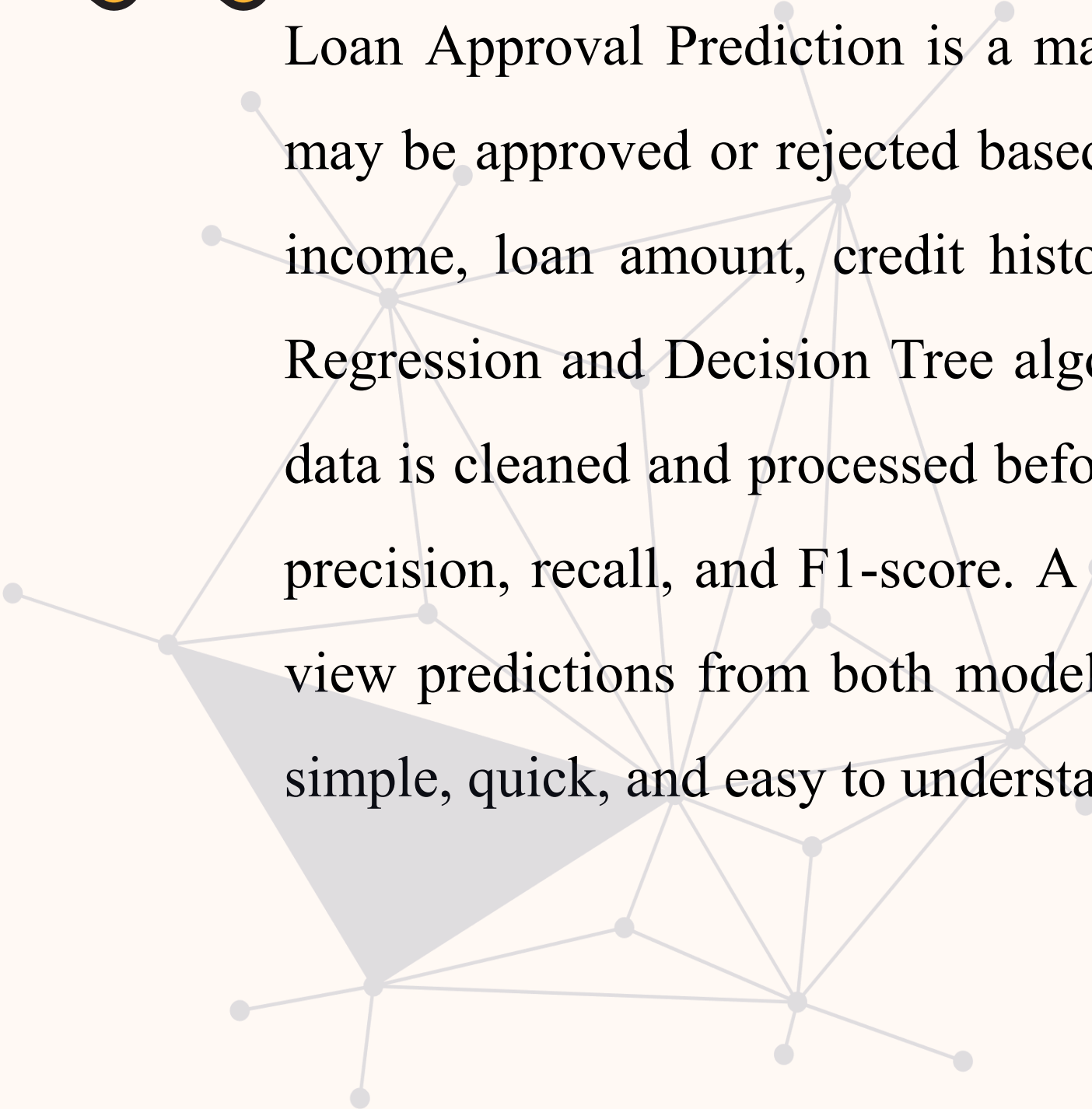
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LOAN
APPROVAL
PREDICTION

ABSTRACT



Loan Approval Prediction is a machine learning project that predicts whether a loan application may be approved or rejected based on applicant and loan details. The system uses factors such as income, loan amount, credit history, education, employment status, and property area. Logistic Regression and Decision Tree algorithms are used to train the models and make predictions. The data is cleaned and processed before model training, and the models are evaluated using accuracy, precision, recall, and F1-score. A Streamlit-based interface allows users to enter their details and view predictions from both models. The system aims to make the initial loan prediction process simple, quick, and easy to understand while reducing manual effort.



INTRODUCTION



Loan Approval Prediction is a machine learning project developed to predict whether a loan application may be approved or rejected. The system uses important applicant details such as income, loan amount, credit history, education, employment status, loan term, and property area to make the prediction. The project uses Logistic Regression and Decision Tree algorithms to learn patterns from previous loan application data. Before training, the data is cleaned and processed to make it suitable for the models. A Streamlit-based user interface is also provided, where users can enter their details and receive predictions from both models. The main purpose of the project is to make the initial loan prediction process simple, quick, and easy to understand, while reducing manual effort.

NEED FOR THE STUDY

- To reduce manual effort in the initial loan application analysis.
- To make loan prediction faster and simpler using machine learning.
- To predict loan approval or rejection based on applicant details.
- To compare Logistic Regression and Decision Tree for loan prediction.
- To evaluate the models using accuracy, precision, recall, and F1-score.
- To provide an easy-to-use Streamlit interface for entering details and viewing predictions.



OBJECTIVES

- To create a machine learning system that predicts loan approval or rejection.
- To use Logistic Regression and Decision Tree for prediction.
- To prepare and clean the loan data for training and testing.
- To check model performance using accuracy, precision, recall, and F1-score.
- To get predictions from both models for comparison.
- To create a simple Streamlit interface for entering loan details and viewing results.



PURPOSE OF THE PROJECT

The purpose of this project is to develop a simple machine learning system that can predict whether a loan application may be approved or rejected. It uses Logistic Regression and Decision Tree algorithms to analyze applicant and loan details. The system provides quick predictions through a Streamlit interface, helping reduce manual effort and making the loan prediction process easier to understand.

LITERATURE SURVEY

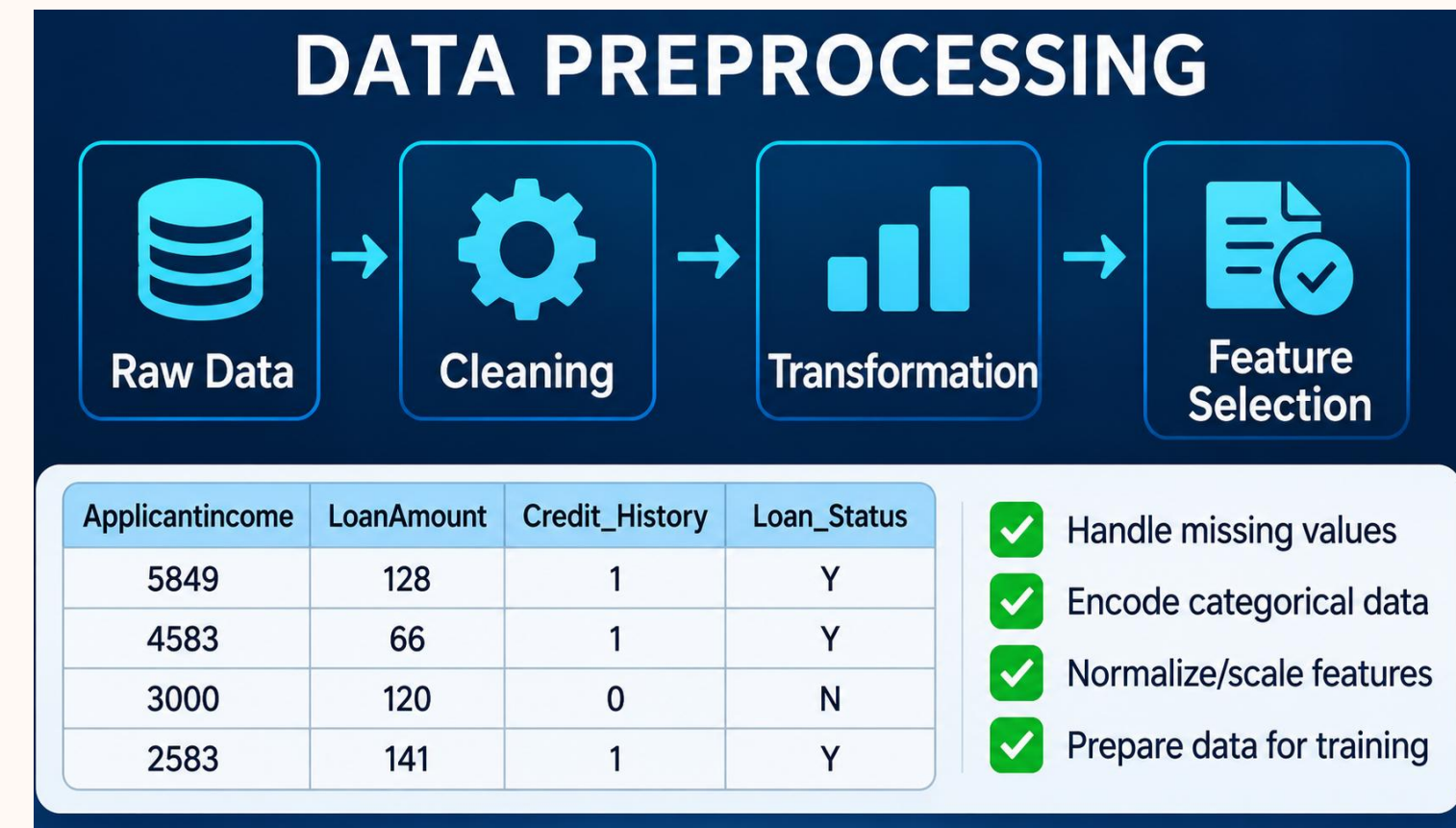
No.	Title	Authors	Year	Results / Findings
1	A Comparative Study of Loan Approval Prediction Using Machine Learning Methods	Vahid Sinap	2024	Achieved 97.71% accuracy . Feature selection using RFE improved the model performance, and cross-validation gave reliable results. (DergiPark)
2	Loan Approval Prediction Model: A Comparative Analysis	Afrah Khan, Eakansh Bhadola, Abhishek Kumar, Nidhi Singh	2021	Compared different ML algorithms. Random Forest showed the strongest predictive performance . (Mili Link)
3	Loan Approval Prediction Based on Machine Learning Approach	Kumar Arun, Garg Ishan, Kaur Sanmeet	2016	Used previous loan records to train and compare ML models. The approach aimed to reduce loan processing time, risk, and manual effort . (IJRASET)

LITERATURE SURVEY

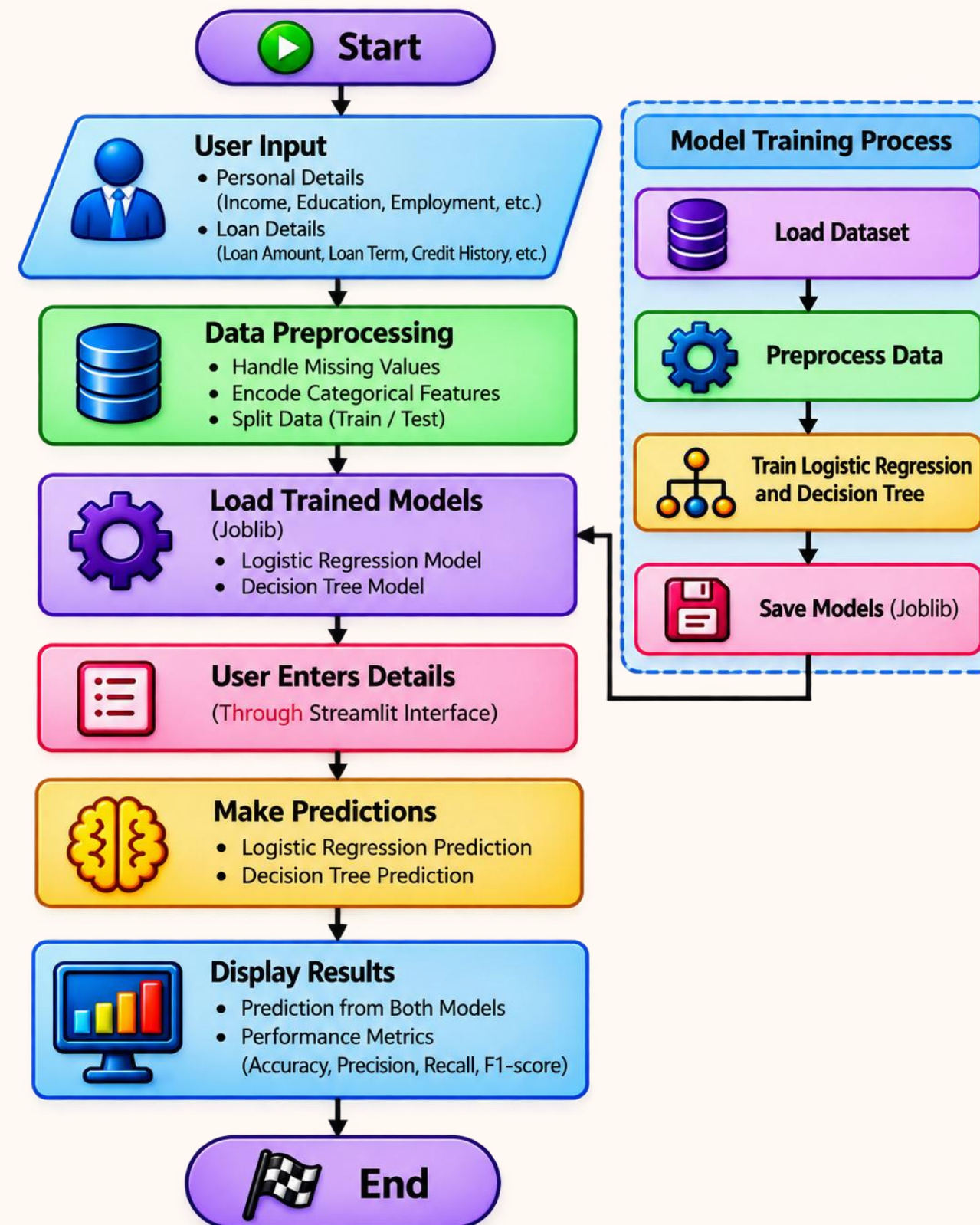
4	Loan Approval Prediction Using Decision Tree Techniques	Yeap Wan Teng, Ch'ng Chee Keong	2025	Decision Tree achieved about 97.8% accuracy and 0.992 AUC with an 80:20 split in the report; the published study describes nearly 99% accuracy across its tested partitions. (ResearchGate)
5	Loan Eligibility Prediction Using Logistics Regression Algorithm	Cornelius Mellino Sarungu	2023	Logistic Regression successfully predicted approval/rejection , but the AUC showed that performance could still be improved. (ResearchGate)
6	Loan Approval Prediction Model: A Comparative Analysis	Afrah Khan, Eakansh Bhadola, Abhishek Kumar, Nidhi Singh	2021	Logistic Regression, Decision Tree and Random Forest were compared. The report notes 93.648% accuracy for Decision Tree , while Random Forest showed better generalization.

GAP IN THE LITERATURE

- Many studies focus on individual machine learning models for loan prediction.
- There is a need to compare different models on the same loan application data.
- Some studies focus mainly on model accuracy rather than a simple user interface.
- There is a need for a system that combines prediction, model evaluation, and user interaction in one application.
- This project addresses these gaps by combining Logistic Regression, Decision Tree, and a Streamlit interface for simple loan prediction.

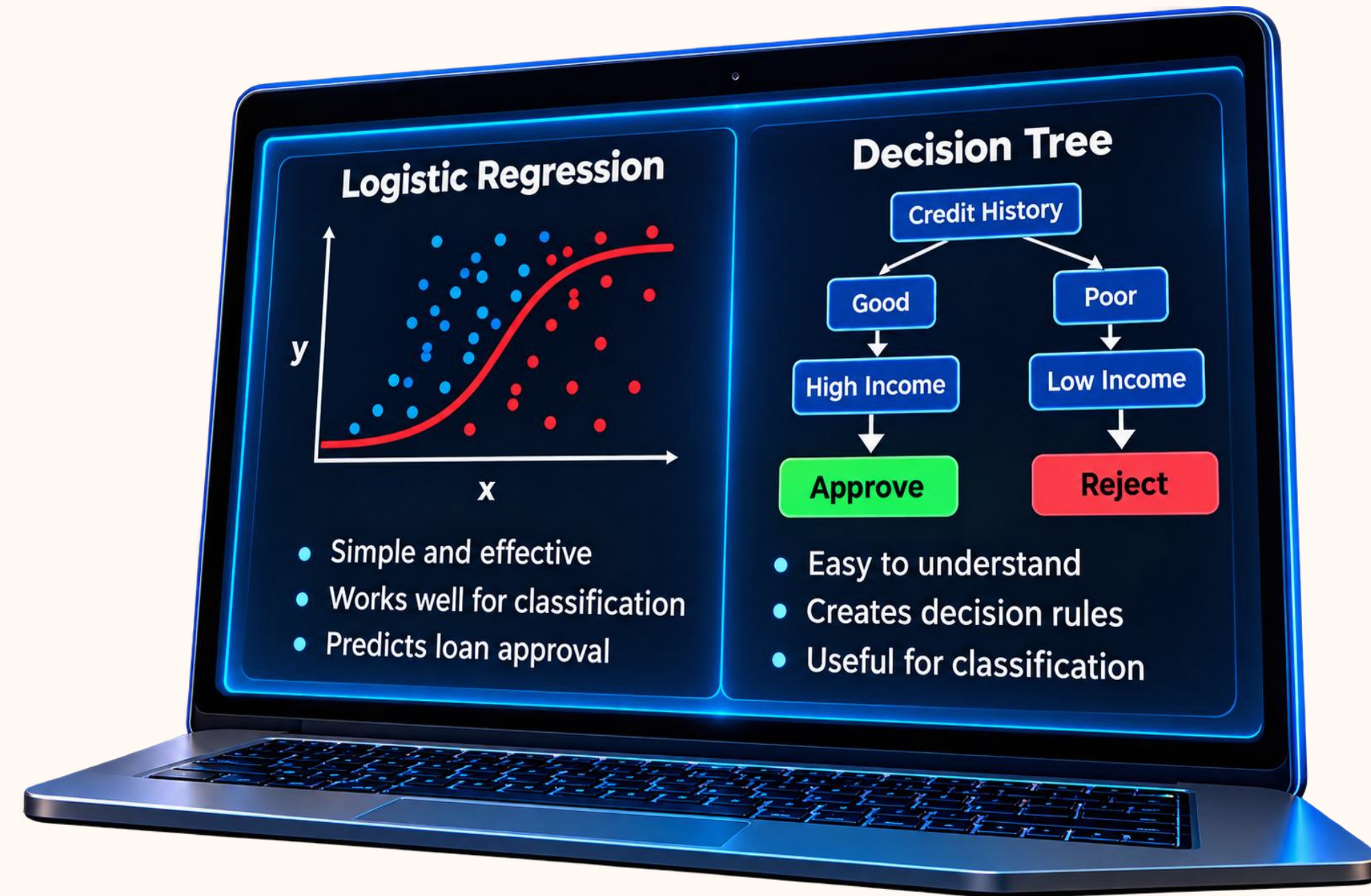


DESIGN



ALGORITHMS USED

- ❖ **Logistic Regression:** Used to predict whether a loan application may be Approved or Rejected based on applicant details.
- ❖ **Decision Tree:** Used to classify loan applications by creating decision rules based on the applicant's information.
- ❖ These two algorithms are trained using the loan dataset and their predictions can be compared using accuracy, precision, recall, and F1-score.



DATASET

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	gender	married	dependents	education	self_employed	applicantincome	coapplicantincome	loanamount	loan_amount	credit_history	property_	loan_status	
2	1	0	0	1	0	5849	0	128	360	1	2	1	
3	1	1	1	1	0	4583	1508	128	360	1	0	0	
4	1	1	0	1	1	3000	0	66	360	1	2	1	
5	1	1	0	0	0	2583	2358	120	360	1	2	1	
6	1	0	0	1	0	6000	0	141	360	1	2	1	
7	1	1	2	1	1	5417	4196	267	360	1	2	1	
8	1	1	0	0	0	2333	1516	95	360	1	2	1	
9	1	1	3	1	0	3036	2504	158	360	0	1	0	
10	1	1	2	1	0	4006	1526	168	360	1	2	1	
11	1	1	1	1	0	12841	10968	349	360	1	1	0	
12	1	1	2	1	0	3200	700	70	360	1	2	1	
13	1	1	2	1	0	2500	1840	109	360	1	2	1	
14	1	1	2	1	0	3073	8106	200	360	1	2	1	
15	1	0	0	1	0	1853	2840	114	360	1	0	0	
16	1	1	2	1	0	1299	1086	17	120	1	2	1	
17	1	0	0	1	0	4950	0	125	360	1	2	1	
18	1	0	1	0	0	3596	0	100	240	1	2	1	
19	1	0	0	1	0	3510	0	76	360	0	2	0	



EXECUTION SCREENSHOT

<<

Model Information

This application uses two supervised classification algorithms.

Logistic Regression

Accuracy

86.99%

F1 Score

91.30%

Deploy

:



Loan Approval Prediction

Predict loan approval using Logistic Regression and Decision Tree



Applicant Details

Enter the applicant's information below and click 'Predict Loan Approval' to obtain predictions from both machine learning models.

Personal Information

Gender

Male

Loan Information

Applicant Income

5000

EXECUTION SCREENSHOT

91.30%

Decision Tree

Accuracy

86.18%

F1 Score

90.81%

The models were trained separately and loaded into this application.

Gender

Male

Married

Yes

Dependents

0

Education

Graduate

Self Employed

No

Applicant Income

5000

Co-applicant Income

2000

Loan Amount

150

Loan Term (months)

360

Credit History

Yes

Property Area

Urban

Deploy



EXECUTION SCREENSHOT

Deploy

Enter the applicant's information below and click 'Predict Loan Approval' to obtain predictions from both machine learning models.

Logistic Regression

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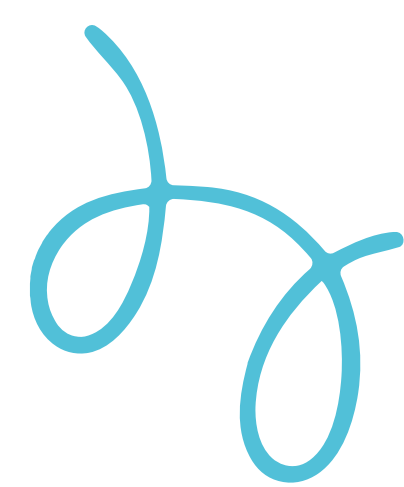
Credit History

Yes

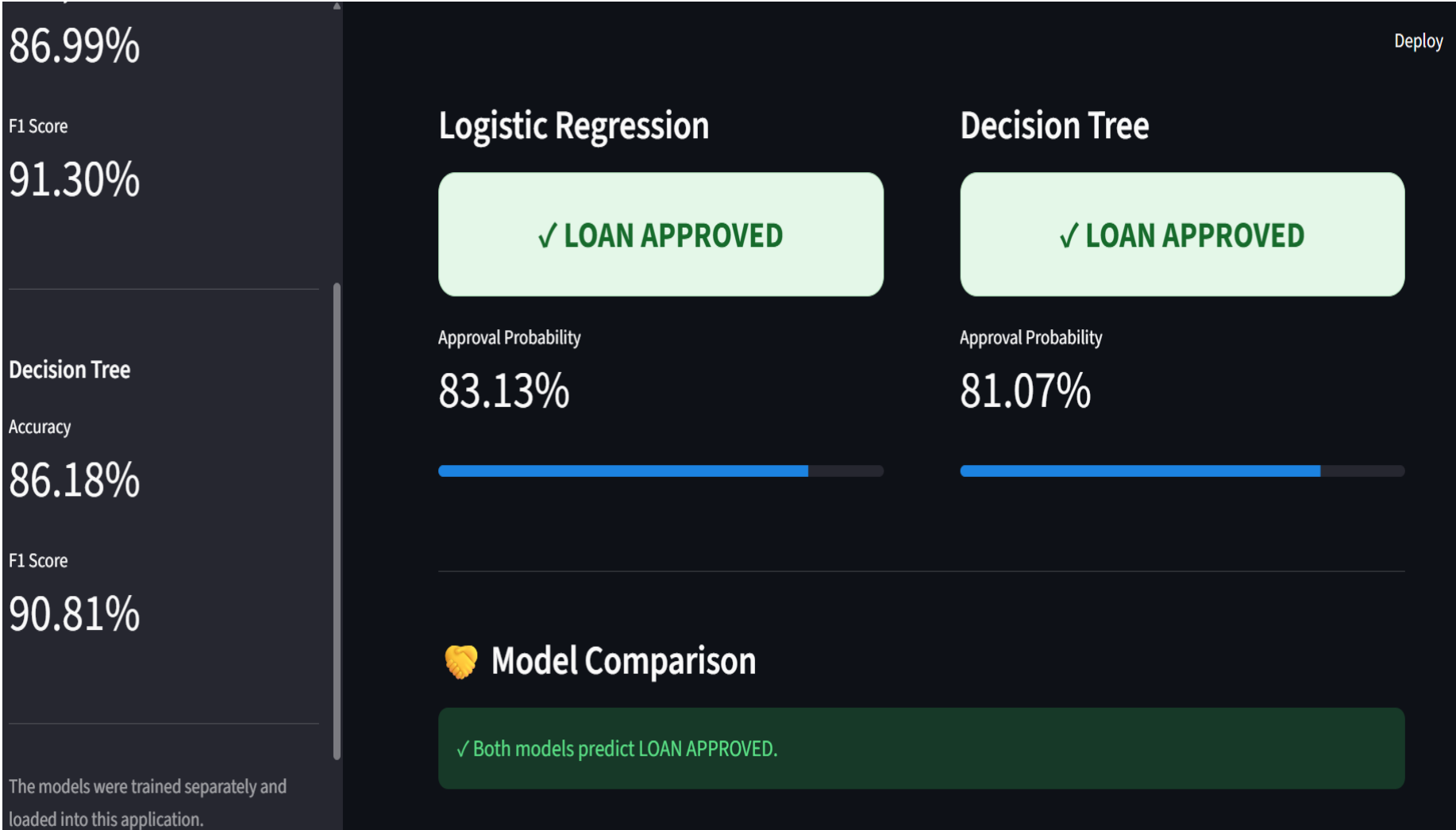
Property Area

Urban

Predict Loan Approval



EXECUTION SCREENSHOT



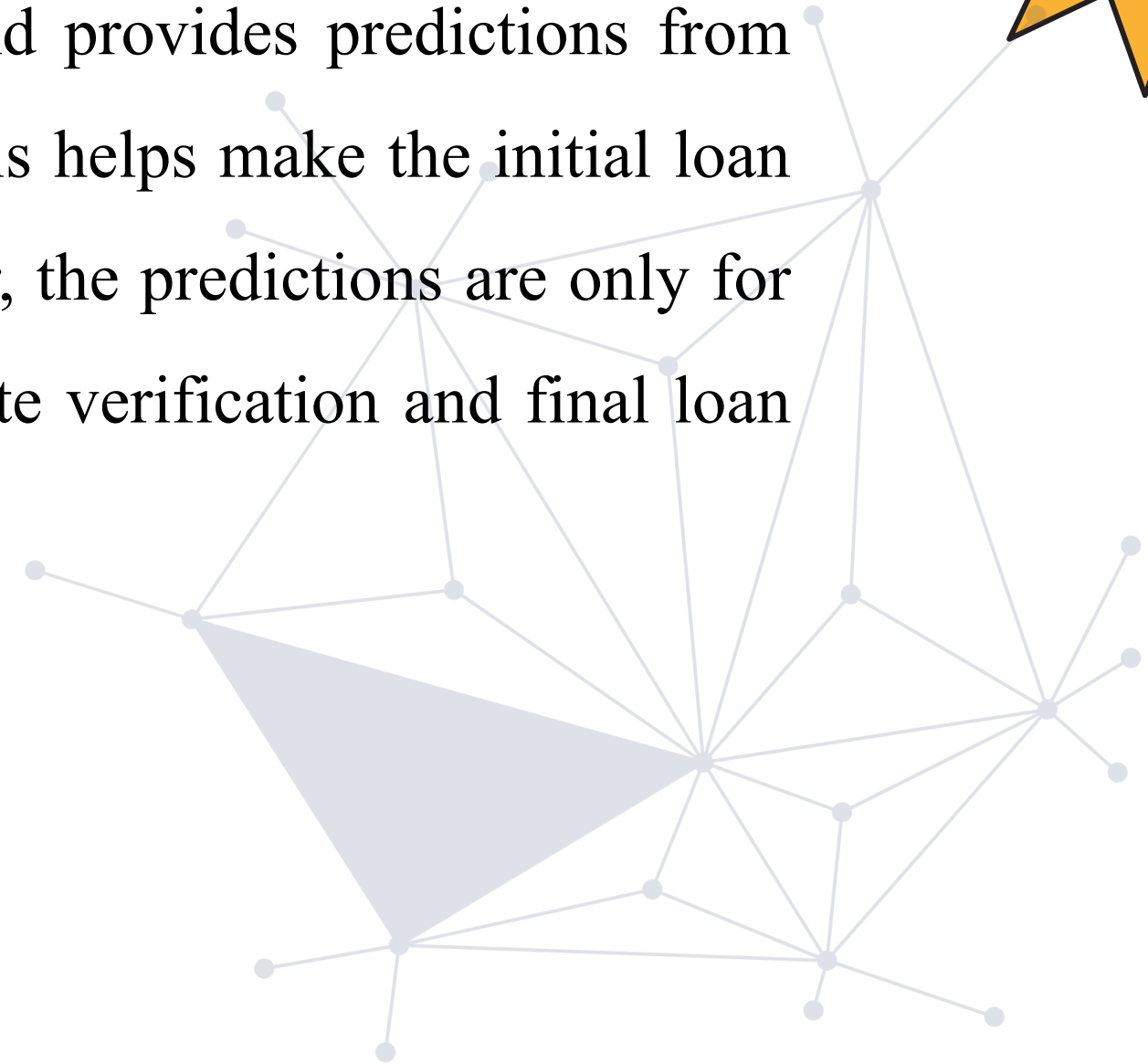
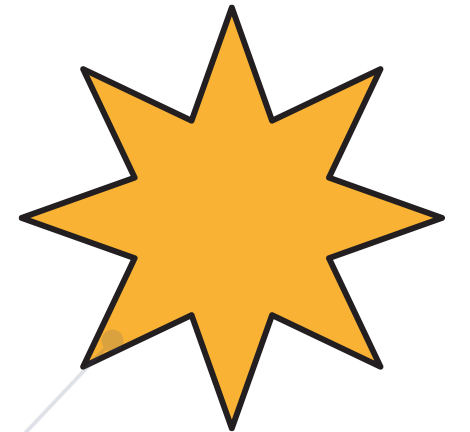
RESULT

The Loan Approval Prediction system successfully predicts whether a loan application may be approved or rejected using Logistic Regression and Decision Tree. The system takes applicant details such as income, loan amount, credit history, education, employment status, and property area as input. The processed data is given to both models, and their prediction results are displayed through the Streamlit interface. The models are evaluated using accuracy, precision, recall, and F1-score. Overall, the project provides a quick, simple, and user-friendly method for initial loan approval prediction.



DISCUSSION

The project demonstrates that Logistic Regression and Decision Tree can be effectively used for loan approval prediction based on applicant and loan-related information. The system processes the input data and provides predictions from both models through a simple Streamlit interface. This helps make the initial loan analysis faster, easier, and more convenient. However, the predictions are only for initial analysis and cannot replace the bank's complete verification and final loan approval process.



CONCLUSION



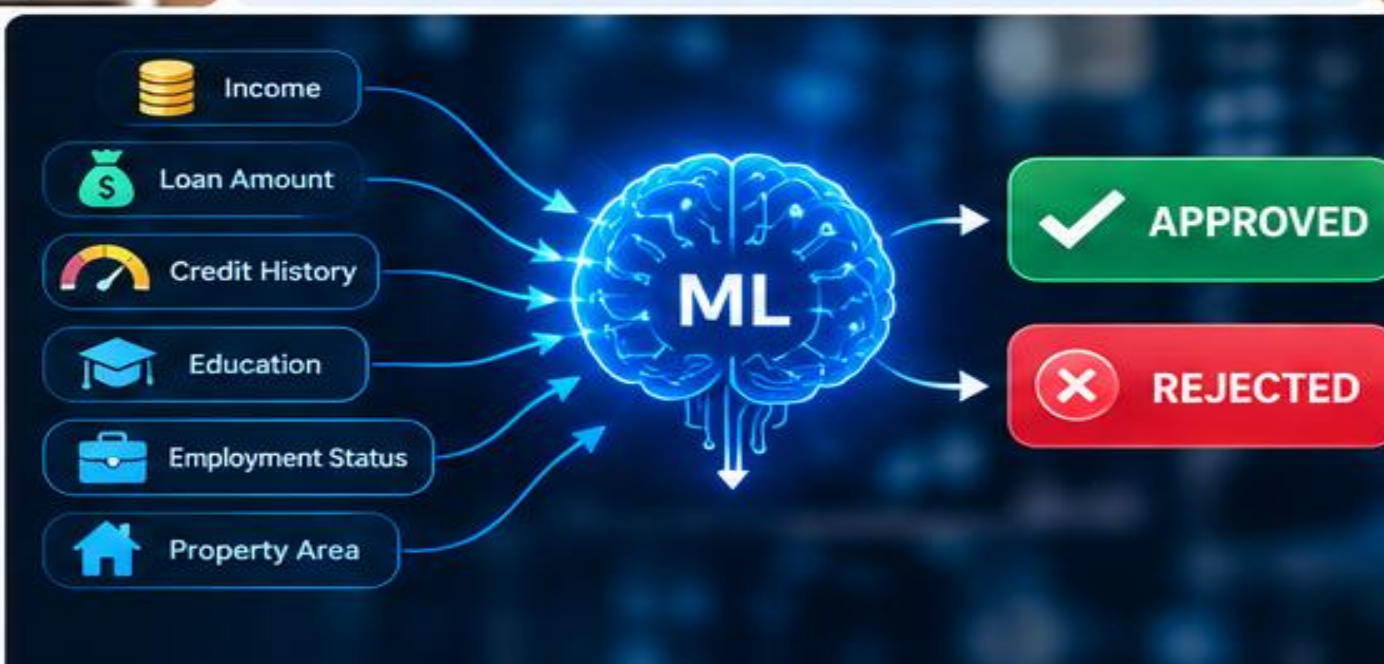
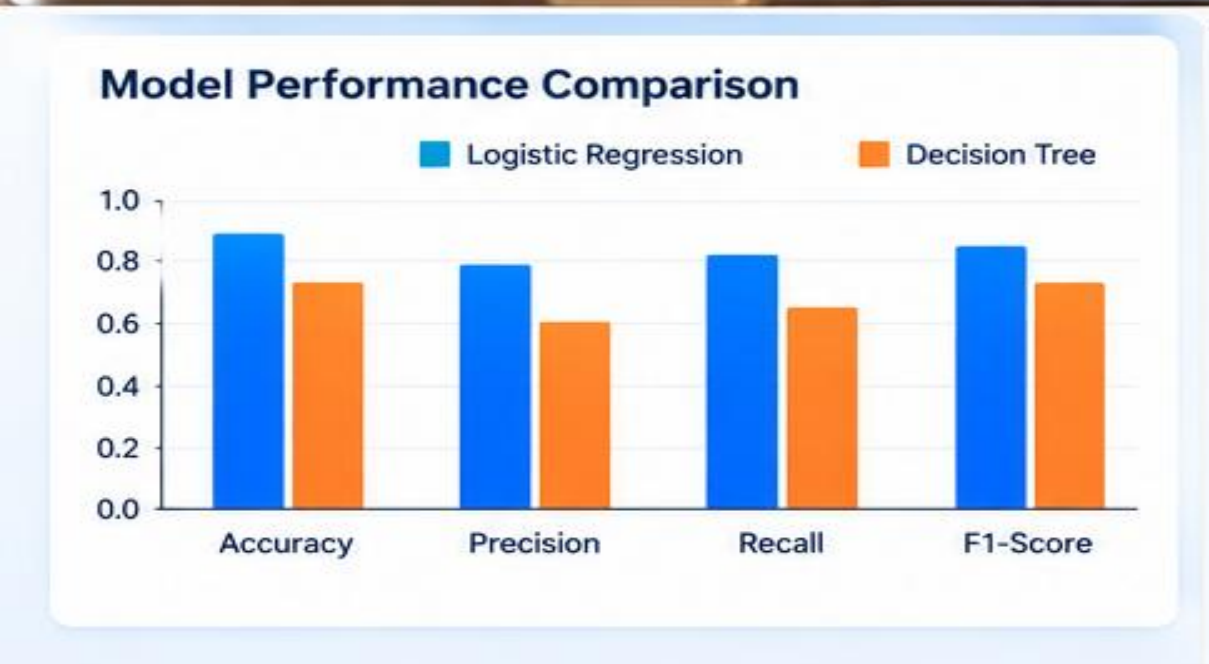
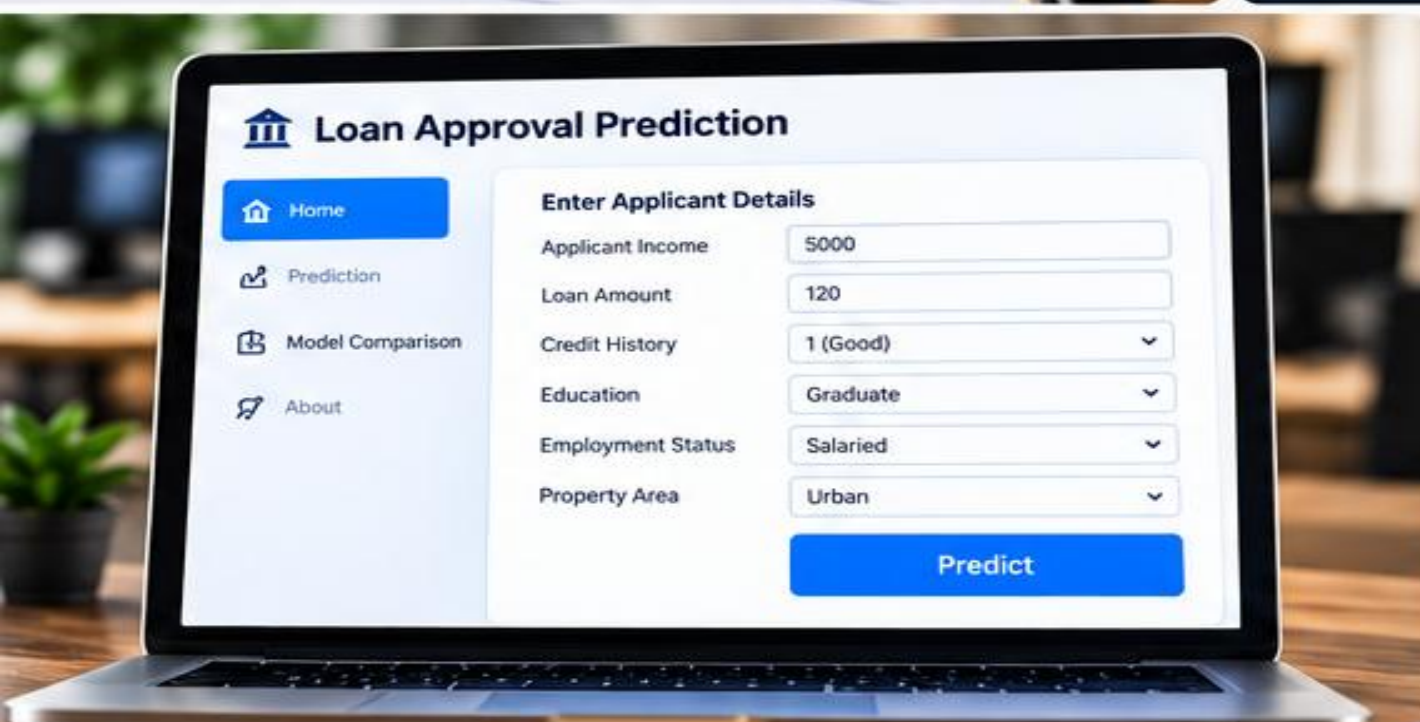
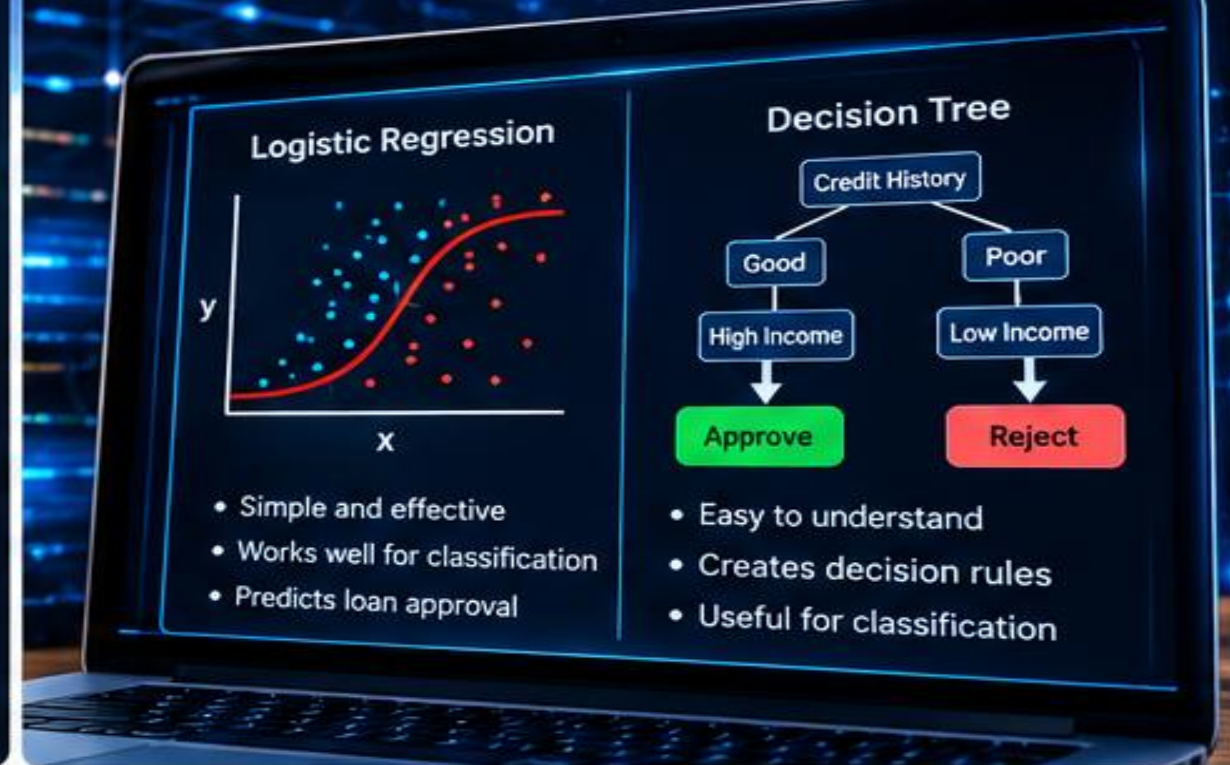
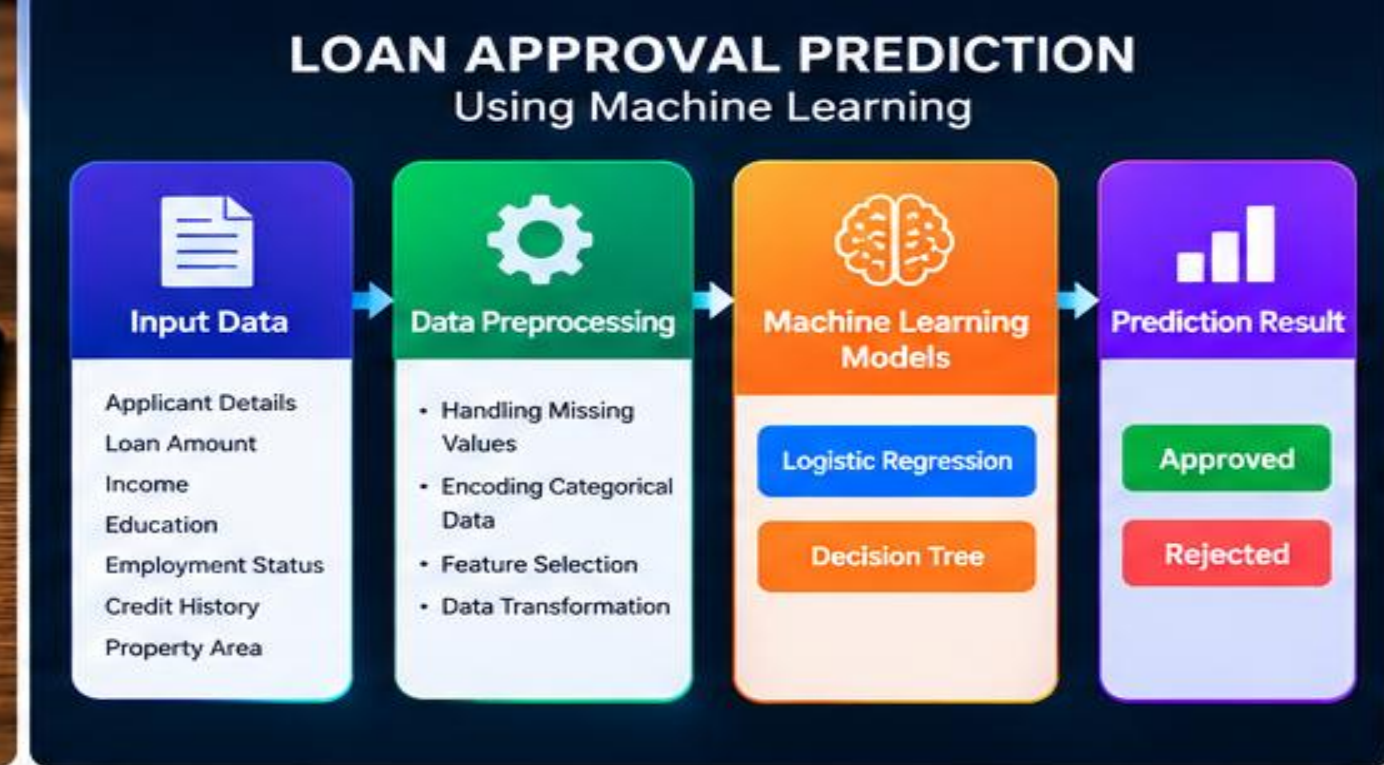
The Loan Approval Prediction project successfully demonstrates the use of machine learning to predict loan approval or rejection based on applicant and loan details. The project uses Logistic Regression and Decision Tree algorithms and provides their predictions through a simple Streamlit interface. It helps make the initial loan prediction process faster, simpler, and easier to understand while reducing manual effort. Overall, the project shows how machine learning can be applied to a real-world loan prediction problem, although the prediction does not replace the bank's final verification and approval process.



FUTURE SCOPE

- Add more machine learning algorithms to improve and compare prediction results.
- Use larger and updated datasets to improve model performance.
- Add more relevant features for better loan prediction.
- Improve the Streamlit interface with more useful and interactive features.
- Regularly update the models when new loan application data becomes available.
- Deploy the system online so it can be accessed easily by users. ML PROJECT





REFERENCES

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2. Afrah Khan, Eakansh Bhadola, Abhishek Kumar, and Nidhi Singh, “Loan Approval Prediction Model: A Comparative Analysis.”
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5. Cornelius Mellino Sarungu, “Loan Eligibility Prediction Using Logistics Regression Algorithm.”
6. Scikit-learn Documentation – Logistic Regression, Decision Tree, model training, and evaluation.
7. Pandas Documentation – Data loading, cleaning, preprocessing, and data manipulation.
8. NumPy Documentation – Numerical calculations and data processing.
9. Streamlit Documentation – Development of interactive Python-based web applications.
10. Joblib Documentation – Saving and loading trained machine learning models. ML PROJECT



THANK
YOU!